Adverse Selection and Queue Priority

Who fills you, when, and what the fill is worth

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The question

A passive fill earns half the spread. It also tells you that someone wanted to trade with you *now*. **How much of the spread is given back to the price move that follows, and what decides it?**

- Information: some traders know more (Glosten-Milgrom, Kyle).
- Queue mechanics: the fill that empties a queue comes right before the price moves through it.
- Speed: stale quotes are picked off in latency races.

In this deck

1. The exact P&L split and markouts.
2. Glosten-Milgrom: the spread as the price of information.
3. Toxicity from queue mechanics alone.
4. The value of queue position.
5. Latency races and toxic flow: evidence.

The accounting identity

Fill j at time t_j (s) and price p_j, with $\varepsilon_j = +1$ if the LP sold and -1 if it bought; inventory $y_t = -\sum_{t_j < t} \varepsilon_j$ (shares); mid S_t. Starting flat, the marked-to-market P&L Π_T is exactly

$$\Pi_T = \underbrace{\sum_j \varepsilon_j (p_j - S_{t_j})}_{\text{spread captured}} + \underbrace{\int_0^T y_{t^-}\, \mathrm{d}S_t}_{\text{inventory P\&L}}.$$

Fill by fill, at horizon h (s), the **markout** is

$$M_j(h) = \varepsilon_j (p_j - S_{t_j+h}) = \underbrace{\varepsilon_j (p_j - S_{t_j})}_{\text{half-spread}} - \underbrace{\varepsilon_j (S_{t_j+h} - S_{t_j})}_{\text{adverse selection}}.$$

- Expected adverse selection is the expected price move against the LP after a fill. Every model in this deck is a model of that term.
- Measurement: choose the benchmark (mid or micro-price), horizons in clock and event time, and errors robust to overlapping horizons.

Glosten-Milgrom: the spread is the price of information

Value $V \in \{V_L, V_H\}$ (price units), belief $\pi = P(V = V_H)$. Each period one trader: informed with probability α (trades in the direction of V), otherwise buys or sells with probability $\frac{1}{2}$. A competitive, risk-neutral market maker sets

$$a = \mathbb{E}[V \mid \text{buy}], \qquad b = \mathbb{E}[V \mid \text{sell}].$$

With $P(\text{buy} \mid H) = \alpha + \frac{1-\alpha}{2}$ and $P(\text{buy} \mid L) = \frac{1-\alpha}{2}$, at $\pi = \frac{1}{2}$:

$$a - b = \alpha(V_H - V_L).$$

After each trade π is updated by Bayes' rule; the spread closes as prices learn V (Glosten and Milgrom, 1985).

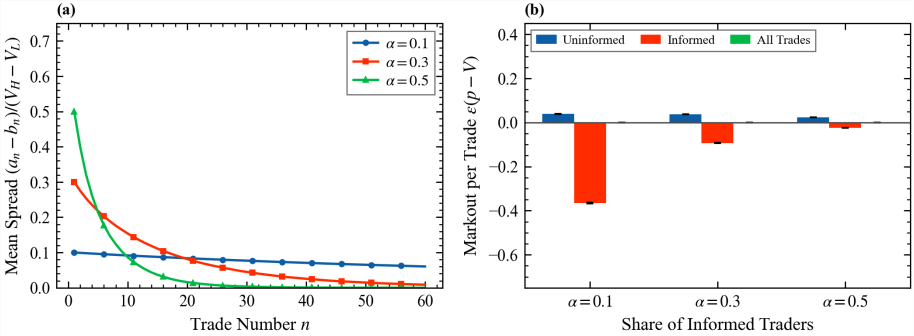
Kyle (1985), one period: noise trades with standard deviation σ_u (shares), value with standard deviation σ_v. In equilibrium the price moves linearly in net order flow, with slope

$$\lambda_K = \frac{\sigma_v}{2\sigma_u}$$

(price per share): impact is the market's inference about information (Kyle, 1985).

Both models: **zero expected profit** for the market maker; gains from uninformed traders pay exactly for losses to informed ones.

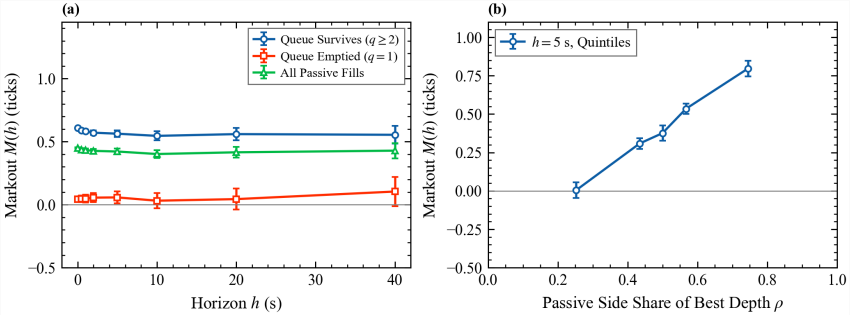
Glosten-Milgrom simulated



(a) 40,000 paths per α, $V_H - V_L = 1$. The first spread is α exactly (0.300 at $\alpha = 0.3$). The mean spread halves after 5 trades at $\alpha = 0.5$, 11 at 0.3, and not within 60 trades at 0.1.

(b) Markout against the true value, per trade, at $\alpha = 0.3$: $+0.0369 \pm 0.0003$ against uninformed traders, -0.0933 ± 0.0005 against informed ones, 0.0001 ± 0.0001 overall (zero profit). Standard errors from path means.

Toxicity without information



Passive fills in the zero-intelligence Markovian book (deck 1): nobody is informed. 16,503 fills in 600,000 events; bars at two standard errors.

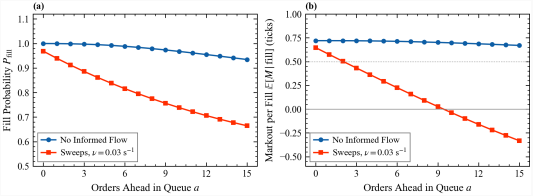
- (a) 0.28 of fills **empty the queue**; they keep 0.04 ± 0.01 ticks against 0.61 ± 0.00 for the others.
- (b) Fills on a thin side keep 0.00 ± 0.03 ticks; on a thick side 0.80 ± 0.03.

Adverse selection is partly mechanical: the fill that takes the last order is the fill that moves the price.

The value of queue position

One sell order at the best ask with a orders ahead and b behind; bid queue c. Rates (s^{-1}): market buys μ hit the front, limit sells λ join the back, each other order cancels at θ, and informed **sweeps** at rate ν take the whole queue and move fair value J ticks (idea from Moallemi and Yuan (2017)).

Value per fill against the mid after the next price change: $\frac{1}{2} - \mathrm{AS}$, with $\mathrm{AS} = 1$ if the fill empties the queue, $2P_{\uparrow}(c, b) - 1$ otherwise, J after a sweep. Solved exactly as an absorbing Markov chain on (a, b, c).



Queue of 16, bid 10, $\nu = 0.03\ \mathrm{s}^{-1}$, $J = 2$. Front: fill probability 0.97, 0.65 ticks per fill. Back: 0.66, -0.33. Joining behind 10 orders loses money; without sweeps the curve is flat.

Latency races and toxic flow: the evidence

Races. London Stock Exchange message data, including failed orders (Aquilina et al., 2022):

- about one race per minute per FTSE 100 symbol, lasting microseconds;
- about 20% of trading volume;
- about a third of price impact and of the effective spread;
- eliminating races would cut the cost of liquidity by about 17%.

Design. Frequent batch auctions replace competition on speed with competition on price (Budish et al., 2015).

Who is toxic. A broker that sees client identities can learn a trend signal from informed flow and choose what to internalize (Cartea and Sánchez-Betancourt, 2025). Neural networks trained by an online Bayesian method predict which client trades are toxic, beating logistic regression and random forests in profit on foreign exchange data (Cartea et al., 2023).

In the models. A race is a sweep: the stale quote is filled exactly when it should not be. Speed buys a lower effective ν in the queue model.

Discussion and takeaways

For a liquidity provider

1. Measure markouts, not fill rates; condition them on queue state, imbalance and counterparty.
2. Priority is an asset. Its value grows with the rate of informed sweeps; joining a long queue late can have negative value.
3. In Glosten-Milgrom the spread is set by α. When informed flow rises, widen or step back; when it falls, competition closes the spread.
4. Some adverse selection is mechanical and predictable from the book alone.

Assumptions. Queue model: one-tick spread, unit orders, sweeps as Poisson events, value measured at the next price change. Glosten-Milgrom: one asset value, unit trades, competitive zero profit.

Open questions

- The equilibrium value of priority when every LP competes for the front.
- Real-time toxicity forecasts with calibrated uncertainty, when toxic traders adapt.
- Measuring ν from market-by-order data.

Reproduce

Code: `adverse_selection.py`, `markouts.py`, `queue_position.py`. Experiments: `e9`, `e10`, `e22`.

```
from stochastic_lob import adverse_selection as gm, queue_position as qp
sim = gm.simulate(alpha=0.3, n_trades=60, n_paths=10_000)
gm.markouts_vs_value(sim)                          # all, uninformed, informed
r = qp.QueueRates(lam=1.0, mu=0.9, theta=0.01, nu=0.03, jump=2.0)
V = qp.solve_queue_value(r, a_max=15)              # fill prob. and value by (a, b, c)
```

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